MINI PROJECT

E-COMMERCE DAILY SALES DATA ANALYSIS AND VISUALIZATION USING PYTHON, POWER BI, AND TABLEAU

AD23632- Framework for Data Visual Analytics

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ABSTRACT

This mini project, carried out for the subject Framework of Data Visualization and Analytics (FDVA), analyses e-commerce sales data using Python, Microsoft Power BI and Tableau. Two datasets are used. The first, ecommerce_orders_raw.csv, holds 250 order records with 22 columns, such as order date, customer, state, category, sub-category, quantity, amount, profit, payment mode and delivery details, covering January to March 2019 in India. The second, retail_daily_sales_raw.csv, holds 3,960 invoice lines with 21 columns, such as invoice number, product description, quantity, unit price, country, product category, customer type and promotion flag, covering 2 to 9 December 2011 in Europe. Both raw files had mixed date formats, numbers stored as text with currency symbols or units, inconsistent spellings, duplicate rows and missing values. Python with Pandas was used to correct these problems and to create new columns such as AOV, Profit_Margin_Pct, Sales, Hour and Day_Name. The cleaned files have 240 rows with 27 columns and 3,948 rows with 25 columns. Matplotlib and Seaborn were then used to create eight figures, including top states, quantity share by category and payment mode, profit by month, hourly sales and a weekly trend. Interactive dashboards called the E-commerce Daily Sales Dashboard were built in Power BI and Tableau, with KPI cards, quarter buttons and charts by state, category, customer and payment mode. The analysis showed total sales of ₹161K with a profit of ₹26K, Maharashtra as the leading state, Clothing as 63% of the quantity sold and Cash on Delivery as the most used payment mode (43.58%). On 9 December 2011 the Tableau dashboard reported sales of $18,737, with Germany as the top country. The project shows how preprocessing, charts and dashboards turn raw sales records into clear information that can support business decisions.

1. Data Preprocessing

Two raw datasets were prepared in Python using Pandas. Part A is the Power BI dataset (ecommerce_orders_raw.csv, 250 rows and 22 columns) and Part B is the Tableau dataset (retail_daily_sales_raw.csv, 3,960 rows and 21 columns). The same steps were applied to both files, as shown below.

Part A: Power BI dataset (ecommerce_orders_raw.csv)

Loading and inspecting the data

```
pb = pd.read_csv("ecommerce_orders_raw.csv")
pb.shape   # (250, 22)
pb.info()
```

Quantity, Amount, Profit and Order_Date were stored as text and had to be converted. Category, state and payment values also used inconsistent spelling.

Missing values and duplicates

```
print(pb.isnull().sum())
print("Duplicate rows:", pb.duplicated().sum())
pb = pb.drop_duplicates().reset_index(drop=True)
```

Age (12), Delivery_Days (15) and Customer_Rating (29) had missing values. 10 duplicate rows were found and removed, which left 240 rows.

Fixing data types and filling missing values

```
pb["Quantity"] = pd.to_numeric(
    pb["Quantity"].astype(str).str.replace(
    r"\s*units?\s*$", "", regex=True, case=False), errors="coerce")
for col in ["Amount", "Profit"]:
    pb[col] = pd.to_numeric(pb[col].astype(str)
              .str.replace(r"[^\d-]", "", regex=True), errors="coerce")
for col in ["Age", "Delivery_Days", "Customer_Rating"]:
    pb[col] = pb[col].fillna(pb[col].median())
```

The word "units" and the symbols ₹ and Rs. with commas were removed, the columns were converted to numbers (the minus sign was kept for losses), and the missing values were filled with the median. After this, every column had 0 missing values.

Standardising text values

```
for col in ["Category","Sub_Category","State","Device_Type",
            "Shipping_Mode","Return_Status","Order_Priority"]:
    pb[col] = pb[col].astype(str).str.strip().str.title()
pm = pb["PaymentMode"].astype(str).str.strip().str.title()
pb["PaymentMode"] = pm.replace({"Cod":"COD","Upi":"UPI",
    "Emi":"EMI","Cash On Delivery":"COD"})
pb["Gender"] = pb["Gender"].astype(str).str.strip().str.upper()
cu = pb["Coupon_Used"].astype(str).str.strip().str.upper()
pb["Coupon_Used"] = cu.replace(
    {"Y":"YES","N":"NO","1":"YES","0":"NO"})
```

The same answer was written in many ways (for example male, Male and MALE, or CHAIRS and Chairs). These were converted to one form. After cleaning, Category has 3 values (Clothing, Electronics, Furniture), PaymentMode has 6 values and Return_Status has 2 values.

Converting the order date

```
def parse_date(x):
    for f in ("%d-%m-%Y","%Y/%m/%d","%d %b %Y","%B %d, %Y"):
        try: return pd.to_datetime(x, format=f)
        except ValueError: continue
    return pd.NaT
pb["Order_Date"] = raw_dates.apply(parse_date)
```

Reading the dates with a single format left 180 dates missing. The dates were written in four formats, so each format was tried in turn. After this, no date was missing (0), and the orders range from 1 January 2019 to 31 March 2019.

New columns and export

```
pb["Order_Month"] = pb["Order_Date"].dt.month
pb["Order_Month_Name"] = pb["Order_Date"].dt.month_name()
pb["Order_Quarter"] = "Qtr " + pb["Order_Date"].dt.quarter.astype(str)
pb["AOV"] = pb["Amount"] / pb["Quantity"]
pb["Profit_Margin_Pct"] = pb["Profit"] / pb["Amount"] * 100
pb.to_csv("ecommerce_orders_cleaned.csv", index=False)   # (240, 27)
```

Five columns were added (Order_Month, Order_Month_Name, Order_Quarter, AOV and Profit_Margin_Pct). The cleaned dataset with 240 rows and 27 columns was saved and used for the Power BI dashboard and Figures 2.1 to 2.4.

Part B: Tableau dataset (retail_daily_sales_raw.csv)

Loading, missing values and duplicates

```
tb = pd.read_csv("retail_daily_sales_raw.csv")
tb.shape   # (3960, 21)
print(tb.isnull().sum())
print("Duplicate rows:", tb.duplicated().sum())
tb = tb.drop_duplicates().reset_index(drop=True)
```

Description (118), CustomerID (141), Delivery_Days (316) and Customer_Rating (594) had missing values. 12 duplicate rows were removed, which left 3,948 rows. Quantity, UnitPrice and InvoiceDate were stored as text.

Fixing data types and filling missing values

```
tb["Quantity"] = pd.to_numeric(
    tb["Quantity"].astype(str).str.replace(
    r"\s*pcs\s*$", "", regex=True, case=False), errors="coerce")
tb["UnitPrice"] = pd.to_numeric(tb["UnitPrice"].astype(str)
    .str.replace(r"[^\d.]", "", regex=True), errors="coerce")
for col in ["Delivery_Days", "Customer_Rating"]:
    tb[col] = tb[col].fillna(tb[col].median())
tb["Description"] = tb["Description"].fillna("UNKNOWN")
```

The unit "pcs" and currency symbols were removed and the columns were converted to numbers. Missing delivery days and ratings were filled with the median, and missing

descriptions were marked UNKNOWN. CustomerID was left empty for guest checkouts, so these rows are not counted as customers.

Standardising text values

```
for col in ["Country","Product_Category","Customer_Type","Promotion",
            "Sales_Channel","Shipping_Mode","Order_Priority",
            "Customer_Segment"]:
    tb[col] = tb[col].astype(str).str.strip().str.title()
tb["Country"] = tb["Country"].replace({"Uk":"United Kingdom",
    "U.K.":"United Kingdom","Holland":"Netherlands"})
rf = tb["Return_Flag"].astype(str).str.strip().str.upper()
tb["Return_Flag"] = rf.replace({"Y":"YES","N":"NO"})
```

Country names such as UK, U.K. and United Kingdom were combined, so Country now has 6 values (Belgium, France, Germany, Netherlands, Norway and United Kingdom). Product_Category has 7 values, and Customer_Type, Promotion and Return_Flag have 2 values each.

Converting the invoice date

```
def parse_dt(x):
    for f in ("%m/%d/%Y %H:%M", "%Y-%m-%d %H:%M:%S",
              "%d-%b-%Y %I:%M %p"):
        try: return pd.to_datetime(x, format=f)
        except ValueError: continue
    return pd.NaT
tb["InvoiceDate"] = raw_dates.apply(parse_dt)
```

Reading the dates with a single format left 2,599 dates missing. After trying all three formats, no date was missing (0), and the invoices range from 2 December 2011 to 9 December 2011.

New columns and export

```
tb["Sales"] = tb["Quantity"] * tb["UnitPrice"]
tb["Date"] = tb["InvoiceDate"].dt.normalize()
tb["Hour"] = tb["InvoiceDate"].dt.hour
tb["Day_Name"] = tb["InvoiceDate"].dt.day_name()
tb.to_csv("retail_daily_sales_cleaned.csv", index=False)   # (3948, 25)
```

Four columns were added (Sales, Date, Hour and Day_Name). The cleaned dataset with 3,948 rows and 25 columns was saved and used for the Tableau dashboard and Figures 2.5 to 2.8.

2. Data Visualization Using Python

Eight figures were created from the cleaned data using Pandas, Matplotlib and Seaborn (sns.set_theme(style="whitegrid")). Figures 2.1 to 2.4 recreate the Power BI dashboard from the Part A data, and Figures 2.5 to 2.8 recreate the Tableau dashboard from the Part B data. Each figure is explained briefly below.

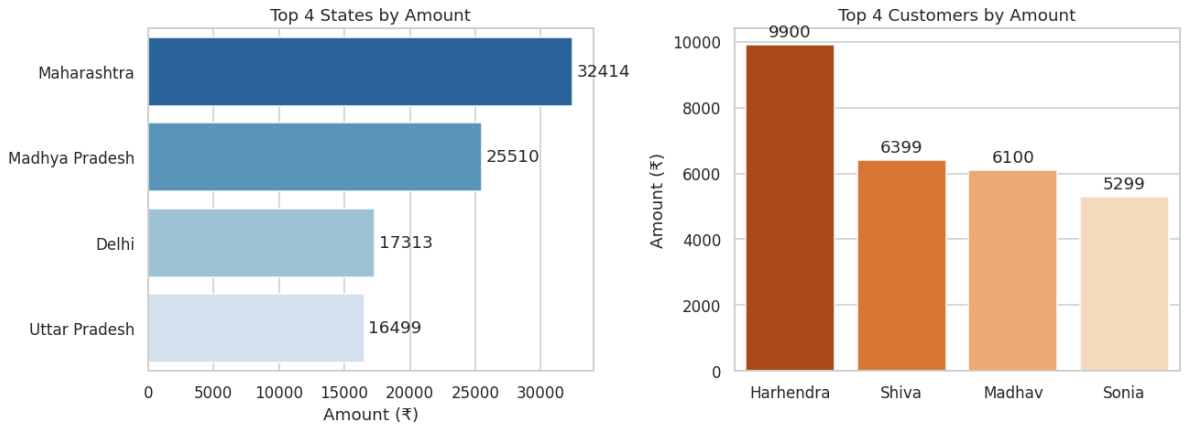


Figure 2.1: Top 4 states and top 4 customers by amount (Power BI data)

Result: Maharashtra leads with ₹32,414, followed by Madhya Pradesh (₹25,510); Harhendra is the top customer (₹9,900).

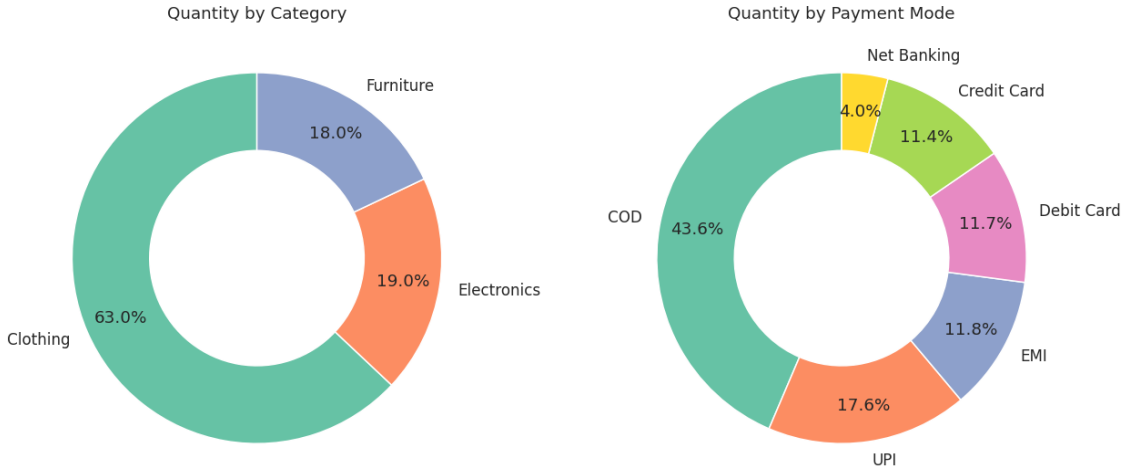


Figure 2.2: Quantity by category and by payment mode (donut charts)

Result: Clothing makes up 63% of the quantity sold, and COD is the most used payment mode (43.58%), followed by UPI (17.58%).

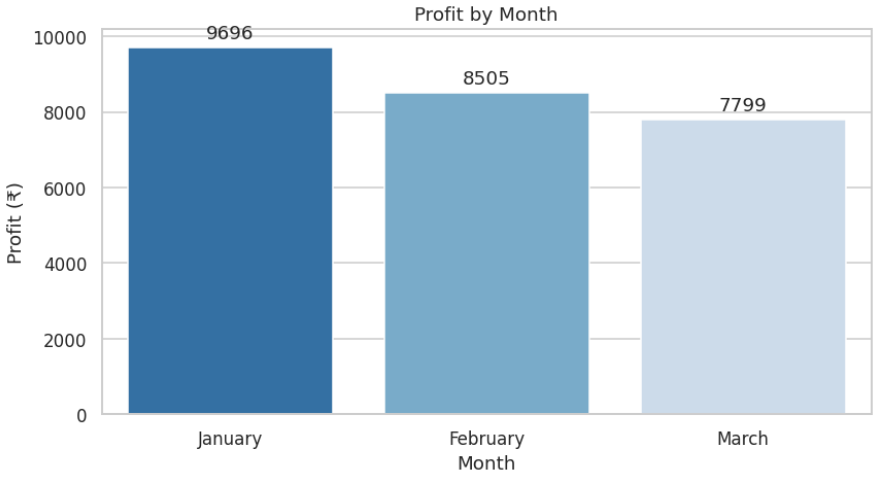


Figure 2.3: Profit by month (bar chart)

Result: Profit falls every month, from ₹9,696 in January to ₹7,799 in March; the total is ₹26,000.

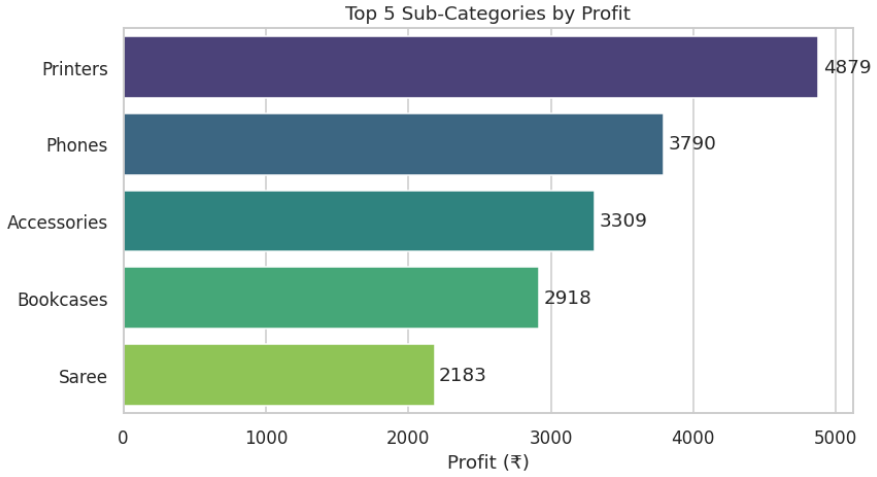


Figure 2.4: Top 5 sub-categories by profit (bar chart)

Result: Printers earns the most profit (₹4,879) and Saree is fifth (₹2,183); the top five together give 66% of all profit.

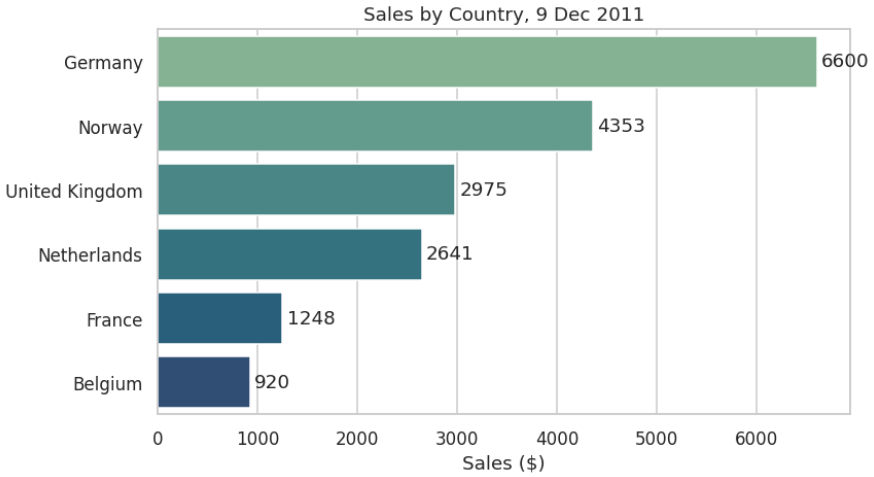


Figure 2.5: Sales by country on 9 December 2011 (Tableau data)

Result: Germany sells the most ($6,600), then Norway ($4,353); Belgium is lowest ($920).

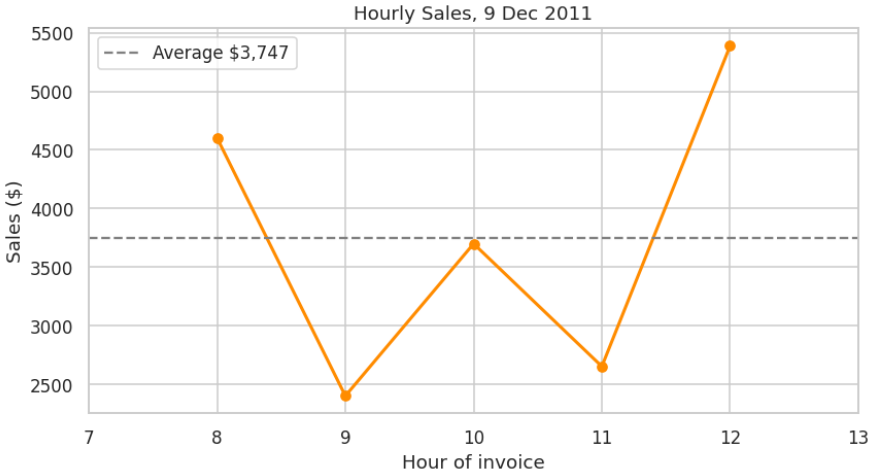


Figure 2.6: Hourly sales on 9 December 2011 (line chart)

Result: Sales peak at 12:00 ($5,388) and are lowest at 9:00 ($2,399); the hourly average is $3,747.

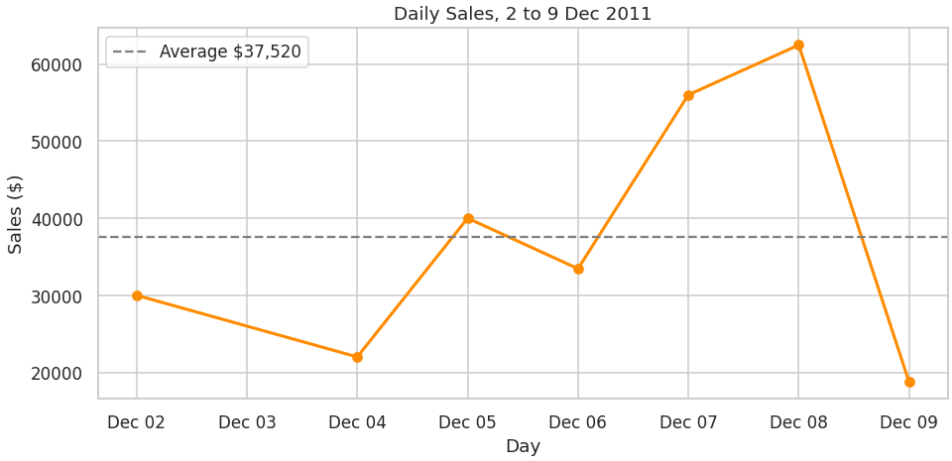


Figure 2.7: Daily sales from 2 to 9 December 2011 (line chart)

Result: Daily sales peak on Dec 08 ($62,458) and the last day, Dec 09, is the lowest ($18,737); there are no sales on Dec 3 (a Saturday).

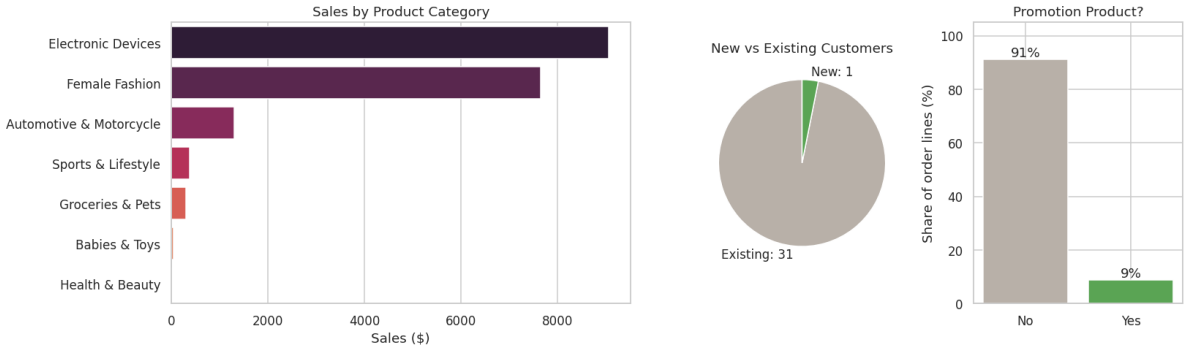


Figure 2.8: Category sales, new vs existing customers and promotion share

Result: Electronic Devices and Female Fashion give 89% of sales; 31 of 32 customers are existing and 91% of lines are not promotion products.

Overall, the Indian orders are led by a few states and by Clothing, which sells the most units, while Cash on Delivery is still the most common way to pay. In the European daily data, sales depend on the hour of the day and on just two product categories, and most buyers are existing customers who buy products that are not on promotion.

3. Power BI Dashboard

Power BI was used to build a one-page interactive dashboard, the "E-commerce Daily Sales Dashboard", from the cleaned orders data (ecommerce_orders_cleaned.csv).

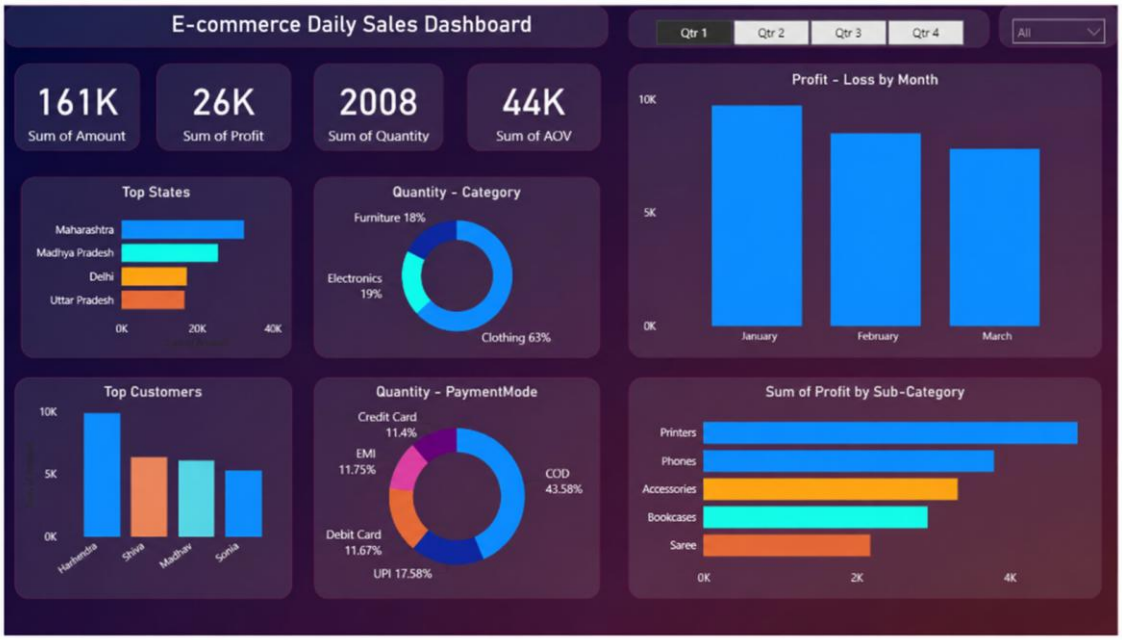


Figure 3.1: E-commerce Daily Sales Dashboard in Power BI

- **Slicers:** quarter buttons Qtr 1 to Qtr 4 and a drop-down list at the top filter the whole page (Qtr 1 is selected, as all orders fall in January to March 2019).
- **KPI cards:** Sum of Amount 161K, Sum of Profit 26K, Sum of Quantity 2008 and Sum of AOV 44K.
- **Charts:** a bar chart of top states, a donut chart of quantity by category, a bar chart of top customers, a donut chart of quantity by payment mode, a bar chart of profit by month and a bar chart of profit by sub-category.
- **Insights:** Maharashtra and Madhya Pradesh lead among the states; Clothing is 63% of the quantity sold; COD is the largest payment mode (43.58%); profit is highest in January and falls in February and March; Printers and Phones give the most profit.
- **Note:** AOV is calculated here as Amount divided by Quantity for each order line, because the card does not state a formula.

4. Tableau Dashboard

Tableau was used to build the "E-commerce Daily Sales Dashboard" for 9 December 2011 from the cleaned retail data (retail_daily_sales_cleaned.csv).

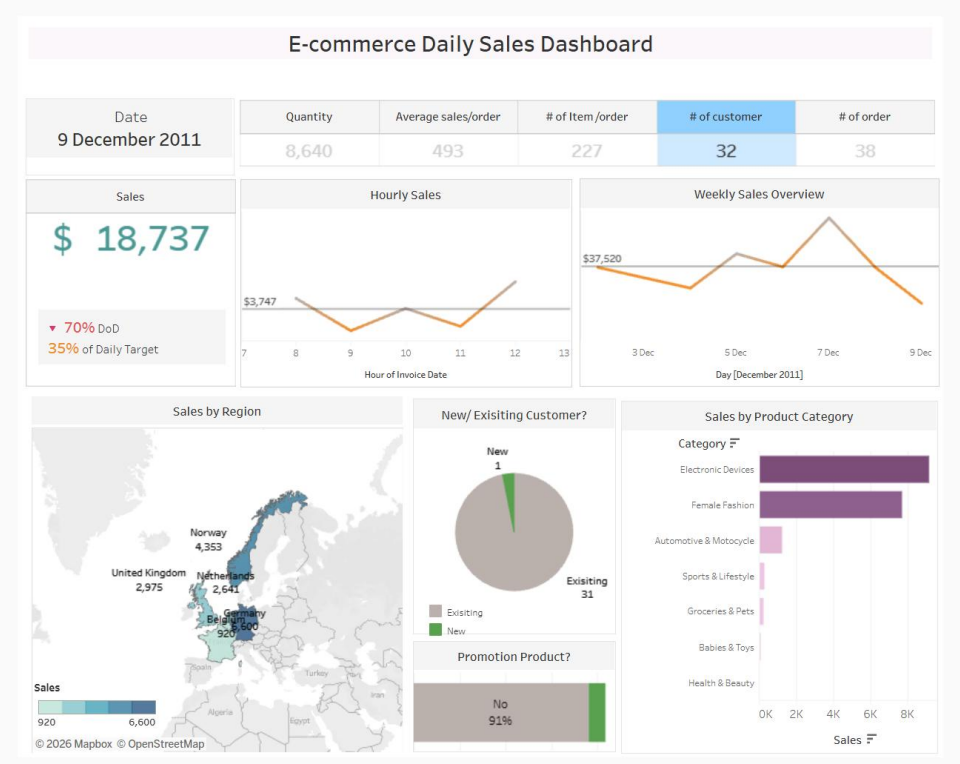


Figure 4.1: E-commerce Daily Sales Dashboard in Tableau

- **Date and KPIs:** the dashboard is set to 9 December 2011 and shows Quantity 8,640, Average sales/order 493, # of Item/order 227, # of customer 32 and # of order 38.
- **Sales card:** Sales of $18,737, which is 70% lower than the previous day (DoD) and 35% of the daily target.
- **Charts:** hourly sales line chart, weekly sales overview line chart, a map of sales by region (country), a pie chart of new and existing customers, a bar of promotion products and a bar chart of sales by product category.
- **Insights:** Germany has the highest sales ($6,600), followed by Norway ($4,353), the United Kingdom ($2,975), the Netherlands ($2,641) and Belgium ($920); 31 of the 32 customers are existing customers; 91% of the sales lines are not promotion products; Electronic Devices and Female Fashion are the largest categories.
- **Note:** the previous-day change and the daily target are taken from the dashboard cards; the weekly chart has no sales on Saturday 3 December.